Table 6-2. Pin Functions—40-Pin DRV8320R Devices (continued)

PIN			TYPE(1)	DESCRIPTION
NAME	NO.			
	DRV8320RH	DRV8320RS		
PGND	1	1	PWR	Device power ground. Also used as the gate drive sink path for the low side MOSFETs. Connect to system ground. PGND mut be connected to AGND and BGND externally.
SCLK	—	23	I	Serial clock input. Serial data is shifted out and captured on the corresponding rising and falling edge on this pin.
SDI	—	22	I	Serial data input. Data is captured on the falling edge of the SCLK pin.
SDO	—	21	OD	Serial data output. Data is shifted out on the rising edge of the SCLK pin. This open drain pin requires an external pullup resistor.
SHA	8	8	I/O	High-side source pin. Connect to the high-side power MOSFET source. This pin is an input for the VDS monitor and the output for the high-side gate driver sink.
SHB	13	13	I/O	High-side source pin. Connect to the high-side power MOSFET source. This pin is an input for the VDS monitor and the output for the high-side gate driver sink.
SHC	16	16	I/O	High-side source pin. Connect to the high-side power MOSFET source. This pin is an input for the VDS monitor and the output for the high-side gate driver sink.
SLA	10	10	I	Low-side source input. Connect to the low-side power MOSFET source. This pin is an input for the VDS monitor.
SLB	11	11	I	Low-side source input. Connect to the low-side power MOSFET source. This pin is an input for the VDS monitor.
SLC	18	18	I	Low-side source input. Connect to the low-side power MOSFET source. This pin is an input for the VDS monitor.
SW	36	36	O	Buck switch node. Connect this pin to an inductor, diode, and the CB bootstrap capacitor.
VCP	4	4	PWR	Charge pump output. Connect an X5R or X7R, 1µF, 25V ceramic capacitor between the VCP and VM pins.
VDRAIN	6	6	I	High-side MOSFET drain input. Connect to the common point of the high-side MOSFET drains. This pin is an input for the VDS monitor.
VDS	23	—	I	VDS monitor trip point setting. This pin is a 7 level input pin set by an external resistor.
VIN	38	38	PWR	Buck regulator power supply input. Place an X5R or X7R, VM-rated ceramic capacitor between the VIN and BGND pins.
VM	5	5	PWR	Gate driver power supply input. Connect to the motor power supply and VDRAIN. Connect an X5R or X7R, 0.1µF, VM-rated ceramic and greater then or equal to 10-uF local capacitance between the VM and PGND pins.
Thermal Pad	PAD	PAD	PWR	Must be connected to ground

(1) PWR = power, I = input, O = output, NC = no connection, OD = open-drain output

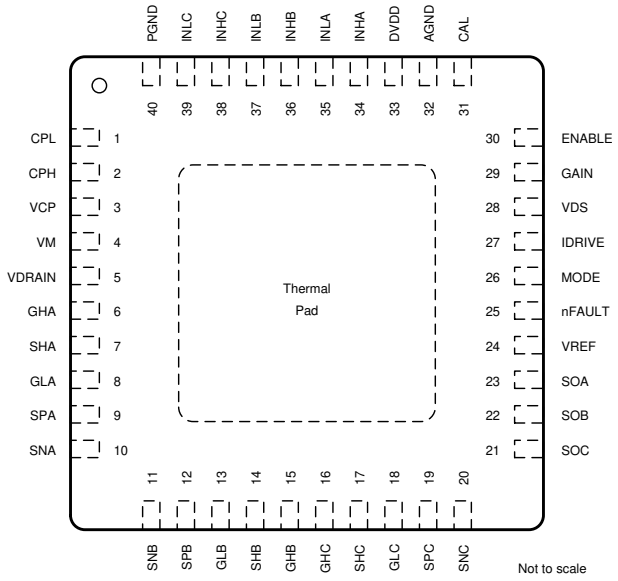


Figure 6-5. DRV8323H RTA Package 40-Pin WQFN With Exposed Thermal Pad Top View

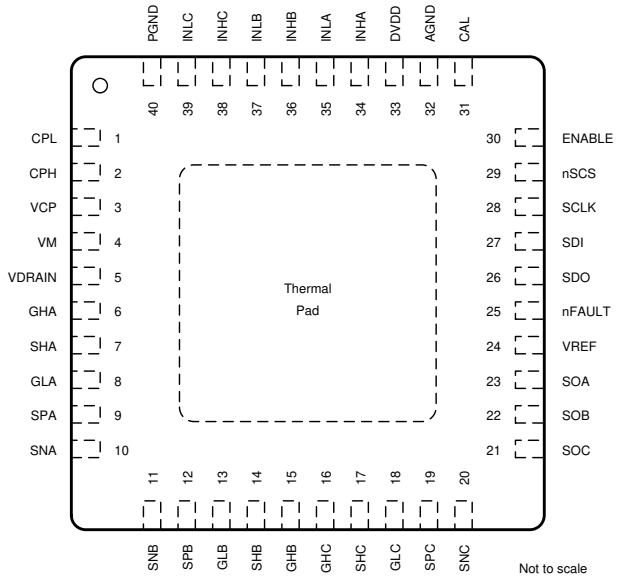


Figure 6-6. DRV8323S RTA Package 40-Pin WQFN With Exposed Thermal Pad Top View

Table 6-3. Pin Functions—40-Pin DRV8323 Devices

PIN			TYPE(1)	DESCRIPTION
NAME	NO.			
	DRV8323H	DRV8323S		
AGND	32	32	PWR	Device analog ground. Connect to system ground. AGND must be connected to PGND externally.

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Table 6-3. Pin Functions—40-Pin DRV8323 Devices (continued)

PIN			TYPE[1]	DESCRIPTION
NAME	NO.			
	DRV8323H	DRV8323S		
CAL	31	31	I	Amplifier calibration input. Set logic high to internally short amplifier inputs and perform auto offset calibration.
CPH	2	2	PWR	Charge pump switching node. Connect an X5R or X7R, 47nF, VM-rated ceramic capacitor between the CPH and CPL pins. For example, in a 24V system the capacitor should be rated 50V (2x margin).
CPL	1	1	PWR	
DVDD	33	33	PWR	3.3V internal regulator output. Connect an X5R or X7R, 1µF, 6.3V ceramic capacitor between the DVDD and AGND pins. This regulator can source up to 30mA externally.
ENABLE	30	30	I	Gate driver enable. When this pin is logic low the device goes to a low-power sleep mode. An 8 to 40µs low pulse can be used to reset fault conditions without entering sleep mode.
GAIN	29	—	I	Amplifier gain setting. The pin is a 4 level input pin set by an external resistor.
GHA	6	6	O	High-side gate driver output. Connect to the gate of the high-side power MOSFET.
GHB	15	15	O	High-side gate driver output. Connect to the gate of the high-side power MOSFET.
GHC	16	16	O	High-side gate driver output. Connect to the gate of the high-side power MOSFET.
GLA	8	8	O	Low-side gate driver output. Connect to the gate of the low-side power MOSFET.
GLB	13	13	O	Low-side gate driver output. Connect to the gate of the low-side power MOSFET.
GLC	18	18	O	Low-side gate driver output. Connect to the gate of the low-side power MOSFET.
IDRIVE	27	—	I	Gate drive output current setting. This pin is a 7 level input pin set by an external resistor.
INHA	34	34	I	High-side gate driver control input. This pin controls the output of the high-side gate driver when using 6x PWM mode.
INHB	36	36	I	High-side gate driver control input. This pin controls the output of the high-side gate driver when using 6x PWM mode.
INHC	38	38	I	High-side gate driver control input. This pin controls the output of the high-side gate driver when using 6x PWM mode.
INLA	35	35	I	Low-side gate driver control input. This pin controls the output of the low-side gate driver when using 6x PWM mode.
INLB	37	37	I	Low-side gate driver control input. This pin controls the output of the low-side gate driver when using 6x PWM mode.
INLC	39	39	I	Low-side gate driver control input. This pin controls the output of the low-side gate driver when using 6x PWM mode.
MODE	26	—	I	PWM input mode setting. This pin is a 4 level input pin set by an external resistor.
nFAULT	25	25	OD	Fault indicator output. This open drain pin pulls logic low during a fault condition and requires an external pullup resistor.
nSCS	—	29	I	Serial chip select. A logic low on this pin enables serial interface communication.
PGND	40	40	PWR	Device power ground. Also used as the gate drive sink path for the low side MOSFETs. Connect to system ground. PGND must be connected to AGND externally.
SCLK	—	28	I	Serial clock input. Serial data is shifted out and captured on the corresponding rising and falling edge on this pin.
SDI	—	27	I	Serial data input. Data is captured on the falling edge of the SCLK pin.
SDO	—	26	OD	Serial data output. Data is shifted out on the rising edge of the SCLK pin. This open drain pin requires an external pullup resistor.
SHA	7	7	I/O	High-side source pin. Connect to the high-side power MOSFET source. This pin is an input for the VDS monitor and the output for the high-side gate driver sink.
SHB	14	14	I/O	High-side source pin. Connect to the high-side power MOSFET source. This pin is an input for the VDS monitor and the output for the high-side gate driver sink.
SHC	17	17	I/O	High-side source pin. Connect to the high-side power MOSFET source. This pin is an input for the VDS monitor and the output for the high-side gate driver sink.
SNA	10	10	I	Current sense amplifier input. Connect to the low-side of the current shunt resistor. If CSA is not used tie to PGND.
SNB	11	11	I	Current sense amplifier input. Connect to the low-side of the current shunt resistor. If CSA is not used tie to PGND.
SNC	20	20	I	Current sense amplifier input. Connect to the low-side of the current shunt resistor. If CSA is not used tie to PGND.
SOA	23	23	O	Current sense amplifier output. If CSA is not used leave this pin disconnected.
SOB	22	22	O	Current sense amplifier output. If CSA is not used leave this pin disconnected.
SOC	21	21	O	Current sense amplifier output. If CSA is not used leave this pin disconnected.
SPA	9	9	I	Low-side current shunt amplifier input. Also VDS monitor negative input for low side MOSFET. Connect to the low-side power MOSFET source and high-side of the current shunt resistor. If CSA is not used tie to SNA.
SPB	12	12	I	Low-side current shunt amplifier input. Also VDS monitor negative input for low side MOSFET. Connect to the low-side power MOSFET source and high-side of the current shunt resistor. If CSA is not used tie to SNB.
SPC	19	19	I	Low-side current shunt amplifier input. Also VDS monitor negative input for low side MOSFET. Connect to the low-side power MOSFET source and high-side of the current shunt resistor. If CSA is not used tie to SNC.
VCP	3	3	PWR	Charge pump output. Connect an X5R or X7R, 1µF, 25V ceramic capacitor between the VCP and VM pins.
VDRAIN	5	5	I	High-side MOSFET drain input. Connect to the common point of the high-side MOSFET drains. This pin is an input for the VDS monitor.
VDS	28	—	I	VDS monitor trip point setting. This pin is a 7 level input pin set by an external resistor.
VM	4	4	PWR	Gate driver power supply input. Connect to the motor power supply and VDRAIN. Connect an X5R or X7R, 0.1µF, VM-rated ceramic and greater then or equal to 10uF local capacitance between the VM and PGND pins.
VREF	24	24	PWR	Current sense amplifier power supply input and reference. Connect an X5R or X7R, 0.1µF, 6.3V ceramic capacitor between the VREF and AGND pins.

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Table 6-3. Pin Functions—40-Pin DRV8323 Devices (continued)

PIN			TYPE[1]	DESCRIPTION
NAME	NO.			
	DRV8323H	DRV8323S		
Thermal Pad	PAD	PAD	PWR	Must be connected to ground

(1) PWR = power, I = input, O = output, NC = no connection, OD = open-drain output

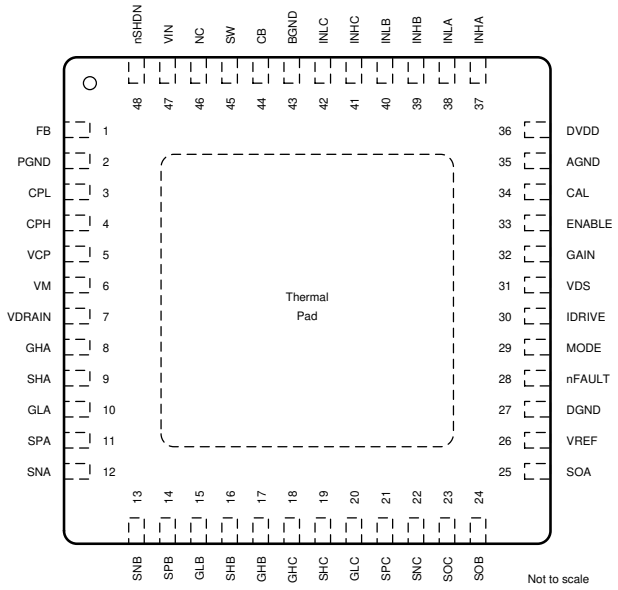


Figure 6-7. DRV8323RH RGZ Package 48-Pin VQFN With Exposed Thermal Pad Top View

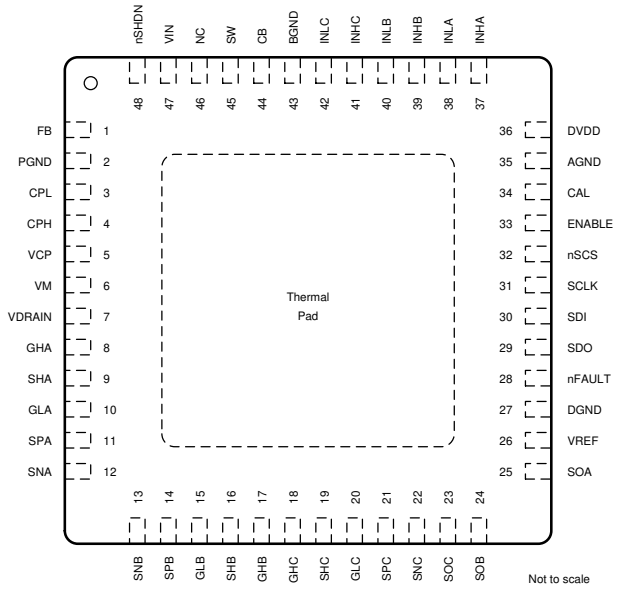


Figure 6-8. DRV8323RS RGZ Package 48-Pin VQFN With Exposed Thermal Pad Top View

Table 6-4. Pin Functions—48-Pin DRV8323R Devices

PIN			TYPE[1]	DESCRIPTION
NAME	NO.			
	DRV8323RH	DRV8323RS		
AGND	35	35	PWR	Device analog ground. Connect to system ground. AGND must be connected to BGND and PGND externally.
BGND	43	43	PWR	Buck regulator ground. Connect to system ground. BGND must be connected to AGND and PGND externally.
CAL	34	34	I	Amplifier calibration input. Set logic high to internally short amplifier inputs and perform auto offset calibration.
CB	44	44	PWR	Buck regulator bootstrap input. Connect an X5R or X7R, 0.1µF, 16V, capacitor between the CB and SW pins.
CPH	4	4	PWR	Charge pump switching node. Connect an X5R or X7R, 47-nF, VM-rated ceramic capacitor between the CPH and CPL pins. For example, in a 24V system the capacitor should be rated 50V (2x margin).
CPL	3	3	PWR	
DGND	27	27	PWR	Device ground. Connect to system ground.
DVDD	36	36	PWR	3.3V internal regulator output. Connect an X5R or X7R, 1µF, 6.3V ceramic capacitor between the DVDD and AGND pins. This regulator can source up to 30mA externally.
ENABLE	33	33	I	Gate driver enable. When this pin is logic low the device goes to a low-power sleep mode. An 8 to 40µs low pulse can be used to reset fault conditions without entering sleep mode.
FB	1	1	I	Buck feedback input. A resistor divider from the buck post inductor output to this pin sets the buck output voltage.
GAIN	32	—	I	Amplifier gain setting. The pin is a 4 level input pin set by an external resistor.
GHA	8	8	O	High-side gate driver output. Connect to the gate of the high-side power MOSFET.
GHB	17	17	O	High-side gate driver output. Connect to the gate of the high-side power MOSFET.
GHC	18	18	O	High-side gate driver output. Connect to the gate of the high-side power MOSFET.
GLA	10	10	O	Low-side gate driver output. Connect to the gate of the low-side power MOSFET.
GLB	15	15	O	Low-side gate driver output. Connect to the gate of the low-side power MOSFET.
GLC	20	20	O	Low-side gate driver output. Connect to the gate of the low-side power MOSFET.
IDRIVE	30	—	I	Gate drive output current setting. This pin is a 7 level input pin set by an external resistor.
INHA	37	37	I	High-side gate driver control input. This pin controls the output of the high-side gate driver when using 6x PWM mode.
INHB	39	39	I	High-side gate driver control input. This pin controls the output of the high-side gate driver when using 6x PWM mode.

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